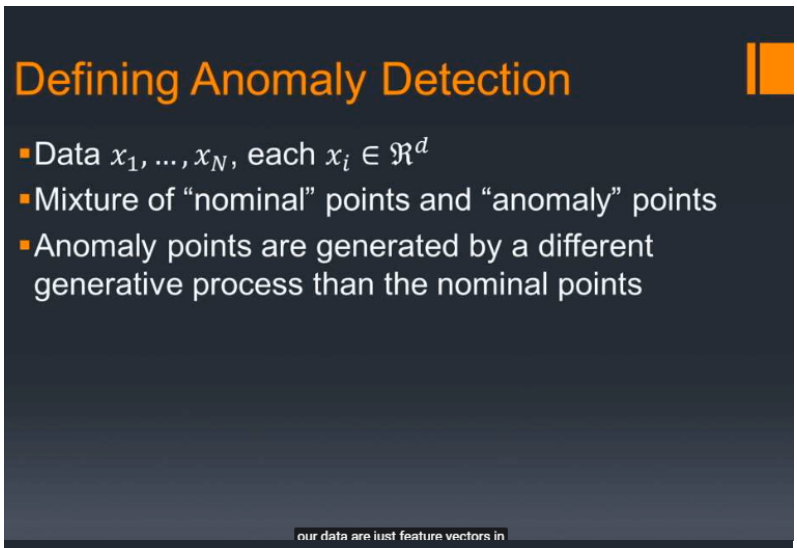
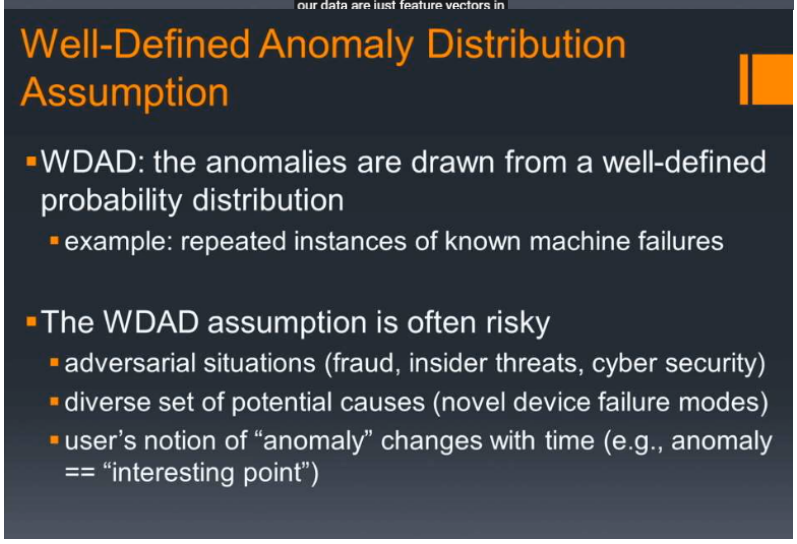
Anomaly Detection: Algorithms, Explanations, Applications - YouTube

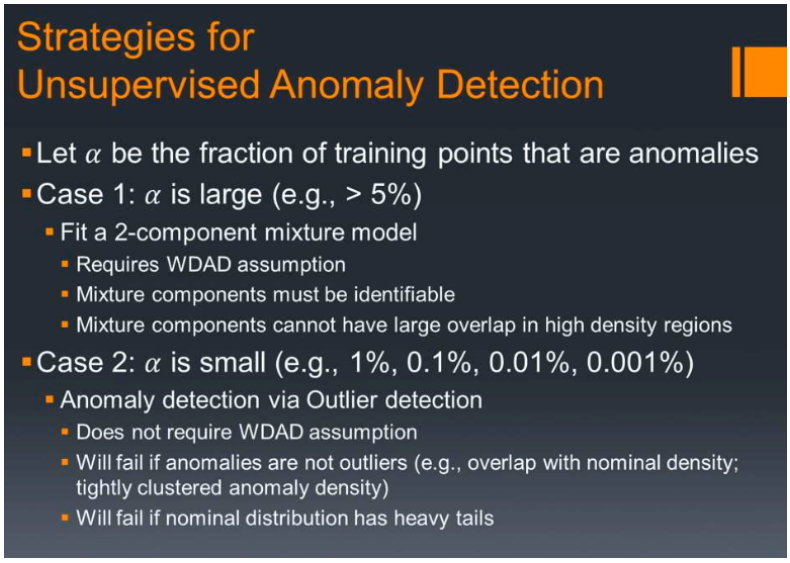
如何定义anomaly，根据anomaly的占比分为两种情况



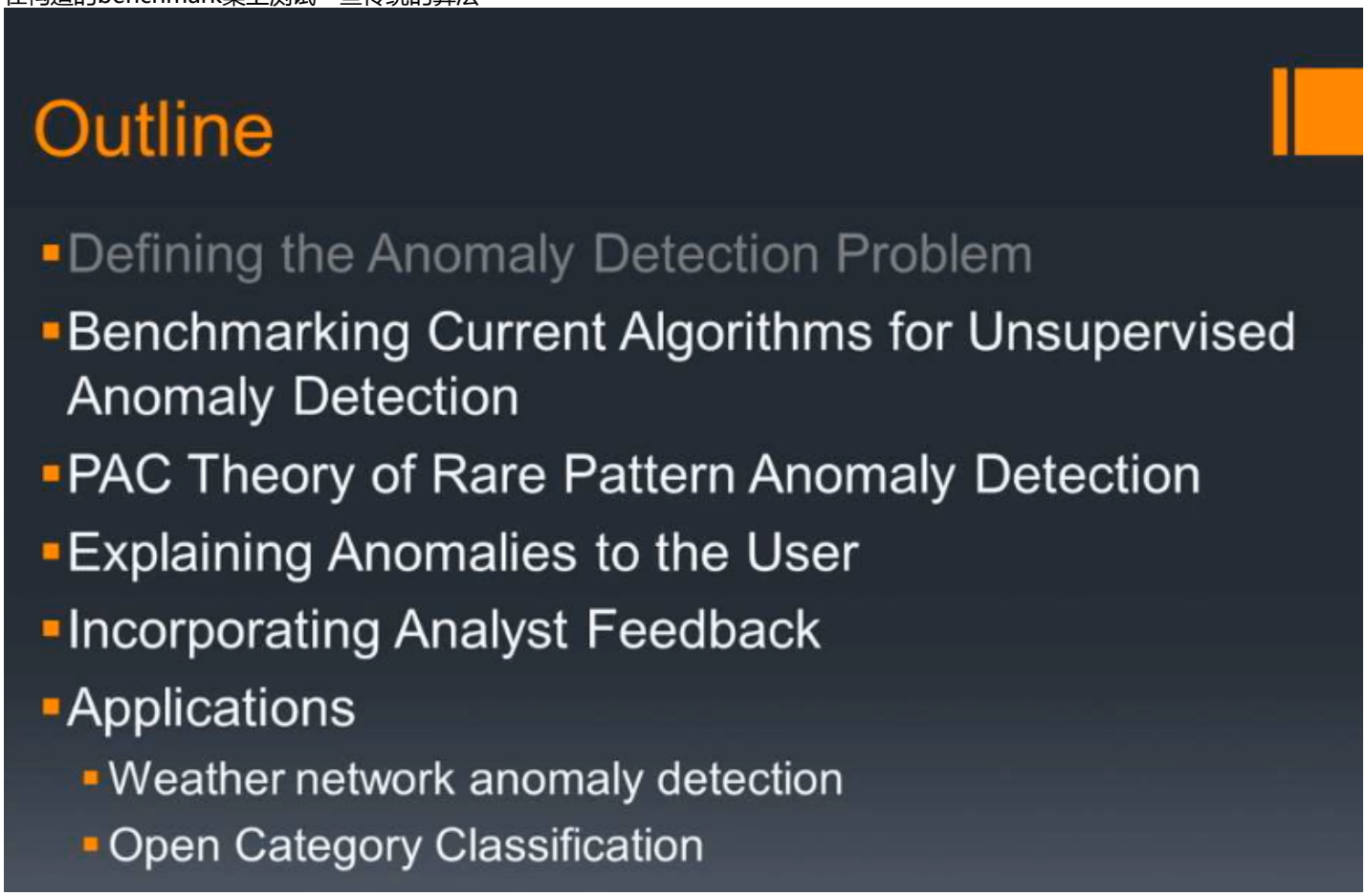


anomaly is a lot => well defined distribution
anomaly is not a lot

1. adversarial
2. not outliers
3. nominal has long tail



在构造的benchmark集上测试一些传统的算法



Benchmarking Study

[Andrew Emmott]

- Most AD papers only evaluate on a few datasets
- Often proprietary or very easy (e.g., KDD 1999)
- Research community needs a large and growing collection of public anomaly benchmarks

[Emmott, Das, Dietterich, Fern, Wong, 2013; KDD ODD-2013]
[Emmott, Das, Dietterich, Fern, Wong. 2016; arXiv 1503.01158v2]

Benchmarking Methodology

- Select 19 data sets from UC Irvine repository
- Choose one or more classes to be “anomalies”; the rest are “nominals”
- Manipulate
  - Relative frequency
  - Point difficulty
  - Irrelevant features
  - Clusteredness
- 20 replicates of each configuration
- Result: 25,685 Benchmark Datasets

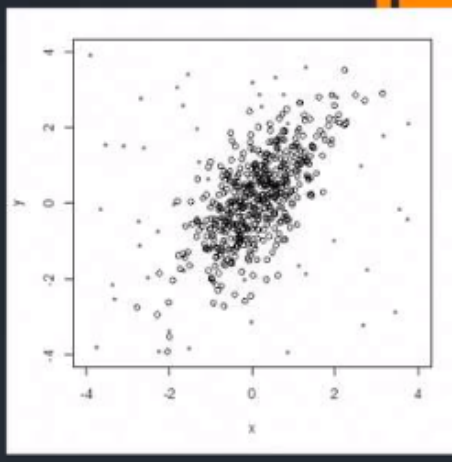
Algorithms

- Density-Based Approaches
  - RKDE: Robust Kernel Density Estimation (Kim & Scott, 2008)
  - EGMM: Ensemble Gaussian Mixture Model (our group)
- Quantile-Based Methods
  - OCSVM: One-class SVM (Schoelkopf, et al., 1999)
  - SVDD: Support Vector Data Description (Tax & Duin, 2004)
- Neighbor-Based Methods
  - LOF: Local Outlier Factor (Breunig, et al., 2000)
  - ABOD: kNN Angle-Based Outlier Detector (Kriegel, et al., 2008)
- Projection-Based Methods
  - IFOR: Isolation Forest (Liu, et al., 2008)
  - LODA: Lightweight Online Detector of Anomalies (Pevny, 2016)

Isolation Forest [Liu, Ting, Zhou, 2011]

- Construct a fully random binary tree
  - choose attribute j at random
  - choose splitting threshold θ uniformly from $[\min(x_{\cdot j}), \max(x_{\cdot j})]$
  - until every data point is in its own leaf
  - let $d(x_i)$ be the depth of point x_i
- repeat 100 times
  - let $\bar{d}(x_i)$ be the average depth of x_i
  - $score(x_i) = 2^{-\left(\frac{\bar{d}(x_i)}{r(x_i)}\right)}$
    - $r(x_i)$ is the expected depth

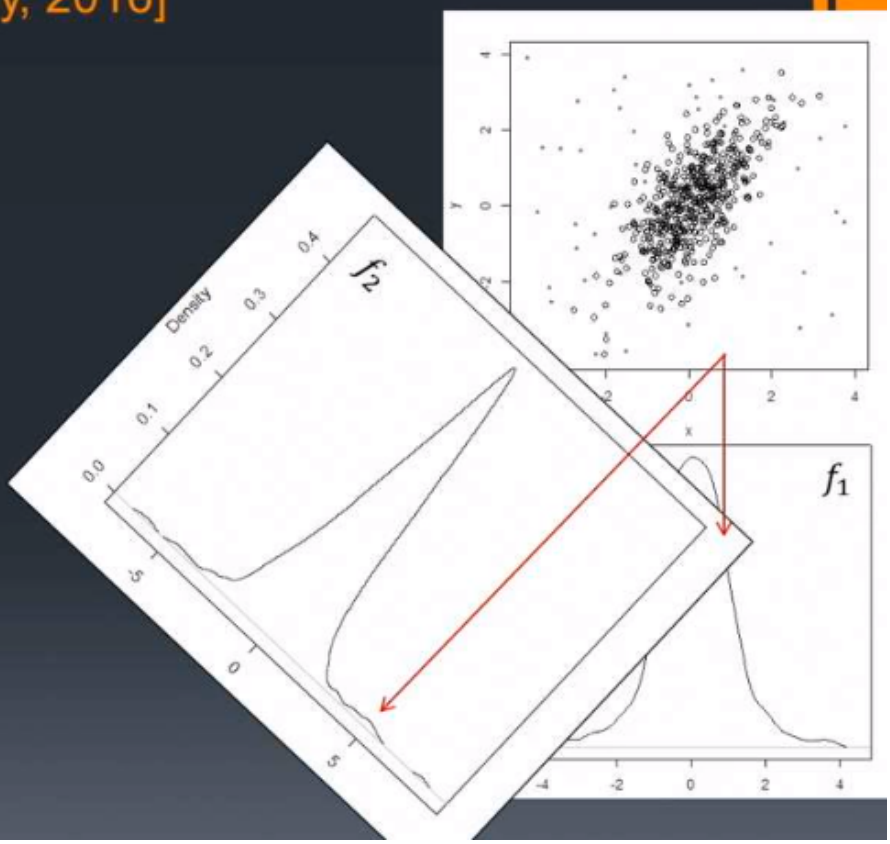
LODA: Lightweight Online Detector of Anomalies [Pevny, 2016]



- $\Pi_1, \ldots, \Pi_M$ set of M sparse random projections
- $f_1, \ldots, f_M$ corresponding 1-dimensional density estimators
- $S(x) = \frac{1}{M}\sum_m -\log f_m(x)$ average "surprise"

LODA: Lightweight Online Detector of Anomalies [Pevny, 2016]

- $\Pi_1, \ldots, \Pi_M$ set of M sparse random projections
- $f_1, \ldots, f_M$ corresponding 1-dimensional density estimators
- $S(x) = \frac{1}{M}\sum_m -\log f_m(x)$ average "surprise"



Pattern Spaces

- A pattern $h: \Re^d \rightarrow \{0,1\}$ is an indicator function for a measurable region in the input space
  - Examples:
    - Half planes
    - Axis-parallel hyper-rectangles in $[-1,1]^d$
- A pattern space $\mathcal{H}$ is a set of patterns (countable or uncountable)

Rare and Common Points

- A point x is τ-rare if there exists a τ-rare h such that $h(x) = 1$
- Otherwise a point is τ-common
- Goal: An anomaly detection algorithm should output all τ-rare points and not output any τ-common points

Algorithm: Greedy Forward Selection

- Anomaly Detector: EGMM
- Feature importance score: Change in log density when feature j is marginalized out

$$score(j) = \log \int_{x_j} P(x_1, \dots, x_j, \dots, x_d)\, dx_j - \log P(x_1, \dots, x_j, \dots, x_d)$$

- $F = \{1, \dots, d\}; Result := []; Model := \text{full model}$
- Repeat
  - Score all $j \in F$, let the highest-scoring feature be j^*
  - $Result := Result + [j^*]$
  - $F := F - \{j^*\}$
  - Update $Model$ to marginalize out j^*
- Until $F == \emptyset$
- Return $Result$